Zero-Noise Extrapolation Study for the Split-Basis H_2O Matrix Entry

27 July 2026

Abstract

This note summarizes the local noisy-simulator zero-noise extrapolation (ZNE) study performed for diagonal split-basis matrix entry $(1, 1)$ of the H_2O calculation at bond length 1.0. It records the simulation and mitigation pipeline, folding factors, shot-allocation analysis, extrapolation models, uncertainty treatment, numerical results, and current interpretation. The main empirical conclusion is that curved extrapolators substantially reduce the observed simulator bias for this relatively noisy/deep entry. The final five-factor experiment gives the smallest observed RMSE for an offset-exponential model, closely followed by a quadratic model, while the linear estimator remains much less variable. Evidence for curvature is strong but does not yet meet the conventional $p < 0.05$ threshold.

1 Question and setup

The target is the diagonal matrix element

$$H_{11}^{\text{ideal}} = -74.9897034653.$$

The basis is the orbital-0 split basis: the original state 0 was replaced by states in fixed orbital sectors, and entry $(1, 1)$ is the dominant split sector studied here.

All experiments reported here were run locally. Circuits cached by the sampling workflow were simulated with Qiskit Aer using the noise model and connectivity of `FakeQuebec`. The simulator was configured with the statevector method, finite shots, and qubit truncation. This is a noisy, shot-based simulation, not an ideal statevector expectation-value calculation. Readout errors were corrected using independently calibrated local assignment matrices before the ZNE fit.

The implementation is primarily in

```
src/mitigation.py
scripts/run_probabilistic_zne_entry.py
scripts/fit_zne_models.py
```

and portable results are under

```
saved/runtime_runs/
  20260724T024719Z_h2o_1.0_state0_orbital0_split_v1_qwc_fake_quebec/
```

2 Noise amplification and execution

2.1 Probabilistic local folding

For an eligible native noisy gate U, a local fold replaces it by

$$U \longmapsto UU^{\dagger}U,$$

which is ideally equivalent to U but executes additional noisy gates. For fractional amplification $\lambda \in [1, 3]$, each eligible gate is folded with probability

$$p_{\text{fold}} = \frac{\lambda - 1}{2}.$$

Different partial folds need not have identical noisy expectation values. Therefore eight independently randomized folding patterns were used for each non-unit factor. The shots assigned to a factor were divided among these patterns; randomization increased the number of circuits, not the total shots at that factor.

The final factor grid was

$$\lambda \in \{1,\ 1.25,\ 1.5,\ 1.75,\ 2\}.$$

Smaller fractional factors were preferred to full factors 3 and 5 because this entry is already noisy and deeper folded circuits can approach noise saturation.

Current implementation limitation. Folding acts on the saved transpiled state-preparation circuits. Measurement basis rotations subsequently introduced by the estimator are not presently folded. Thus λ is an amplification factor for the folded circuit portion, not a guarantee that every physical error source is multiplied uniformly. Realized noisy-gate factors and fold manifests are saved and should be inspected when interpreting the fits.

2.2 Readout mitigation

For each measured physical qubit q, calibration circuits prepare $|0\rangle$ and $|1\rangle$ and estimate the assignment matrix

$$A_q = \begin{pmatrix} p(0|0) & p(0|1) \\ p(1|0) & p(1|1) \end{pmatrix}.$$

The measured distribution is corrected with A_q^{-1}. For a multi-qubit Pauli product, the code assumes local assignment errors and applies the corresponding tensor product of single-qubit inverses. Readout mitigation was applied at every noise factor before extrapolation. It was helpful, but earlier tests showed that readout error was not the dominant bias source.

3 Extrapolation and statistical treatment

3.1 Models

Three variance-weighted models were compared:

$$E_{\text{lin}}(\lambda) = \beta_0 + \beta_1 \lambda, \tag{1}$$

$$E_{\text{quad}}(\lambda) = \beta_0 + \beta_1 \lambda + \beta_2 \lambda^2, \tag{2}$$

$$E_{\text{exp}}(\lambda) = E_0 + s\,\frac{1 - \exp(-k\lambda)}{k}. \tag{3}$$

The last parameterization is an offset exponential written so that $E_{\exp}(0) = E_0$ and it has a stable linear limit as $k \to 0$. In every case the fitted value at $\lambda = 0$ is the ZNE estimate.

If estimates $\widehat{E}_i$ have variances v_i, variance-weighted fitting minimizes

$$\chi^2 = \sum_i \frac{[\widehat{E}_i - E_{\text{model}}(\lambda_i)]^2}{v_i}.$$

For a linear extrapolated estimator

$$\widehat{E}(0) = \sum_i c_i \widehat{E}_i,$$

independent factor variances propagate as

$$\text{Var}[\widehat{E}(0)] = \sum_i c_i^2 v_i.$$

Factor-level parametric bootstrap resampling was additionally used for all three fits and is particularly important for the nonlinear exponential fit. The final results use 10 000 bootstrap samples.

3.2 Shot allocation

For fixed extrapolation coefficients and single-shot standard deviations σ_i, minimizing variance at fixed total shots gives

$$N_i \propto |c_i| \sigma_i.$$

An early equal-allocation run used 147 317 shots at each of $\lambda = \{1, 1.5, 2\}$. A pilot-based allocation then assigned

$$(N_1, N_{1.5}, N_2) = (234\,371,\ 45\,356,\ 162\,224),$$

while preserving the same total of 441 951 shots. The variance-weighted linear coefficients after this reallocation became

$$(c_1, c_{1.5}, c_2) = (1.9889,\ 0.0222,\ -1.0111).$$

This illustrates a self-consistency issue: changing N_i changes v_i, which changes a weighted fit and hence changes c_i. Under a strictly linear model the variance-optimal design tends to place nearly all shots at the two endpoints; an interior point is nevertheless valuable for detecting curvature. A production design should therefore predeclare pilot-derived weights and reserve a minimum diagnostic allocation at interior factors, rather than repeatedly adapting weights to the same production data.

4 Numerical development

4.1 Three-factor linear ZNE

The initial readout-mitigated three-factor run at $\lambda = \{1, 1.5, 2\}$ gave

	bias	variance	RMSE
$\lambda = 1$	2.4949	0.00431	2.4958
equal-allocation linear ZNE	1.3022	0.02198	1.3106
optimized weighted ZNE	1.2134	0.01546	1.2198

The pilot allocation and variance-weighted fit reduced the extrapolated variance by approximately 29.7% and the RMSE by approximately 6.9% relative to the equal-allocation linear run.

4.2 Four-factor model comparison

Adding $\lambda = 1.25$ and using 294 634 shots per factor produced:

model	$E(0)$	bias	variance	RMSE
linear	−73.7650	1.2247	0.00943	1.2285
quadratic	−74.3106	0.6791	0.22591	0.8289
exponential	−74.4352	0.5545	0.52684	0.9134

The curved fits reduced observed simulator bias but amplified statistical variance. With only four factors, the curvature evidence was not conclusive.

4.3 Final five-factor independent-seed comparison

Two independently seeded five-factor runs were inverse-variance pooled. The nominal total was 883 902 shots per factor. Resolved folding-pattern variance was included in the factor variances; therefore nominal shot count alone was not treated as the effective precision. The pooled factor data were

λ	$\widehat{E}(\lambda)$	standard deviation
1.00	−72.509300	0.02681
1.25	−72.133633	0.02773
1.50	−71.774107	0.03487
1.75	−71.453188	0.02890
2.00	−71.160255	0.03185

and the full five-point fits were

model	$E(0)$	bias	variance	standard deviation	RMSE
linear	−73.845122	1.144581	0.00307	0.0554	1.1459
quadratic	−74.321020	0.668683	0.07574	0.2752	0.7231
offset exponential	−74.405365	0.584339	0.14510	0.3809	0.6975

The corresponding fit-quality statistics were

model	χ^2/dof	p-value	AIC
linear	3.129/3	0.372	7.129
quadratic	0.034/2	0.983	6.034
exponential	0.051/2	0.975	6.051

Here

$$\mathrm{AIC} = \chi^2 + 2k$$

penalizes the residual χ^2 by twice the number k of fit parameters. Lower AIC is preferred, but differences smaller than roughly 2 do not meaningfully distinguish models. The quadratic has the lowest AIC, but its advantage over linear is only 1.095.

4.4 Held-out prediction and curvature test

The $\lambda = 1.75$ point was excluded from training and then predicted:

model	prediction	residual	standardized residual
linear	−71.475791	0.022603	0.632
quadratic	−71.454738	0.001550	0.041
exponential	−71.456425	0.003237	0.087

The quadratic predicted the held-out point best, closely followed by the exponential. The linear prediction was nevertheless within 0.63 standard deviations and therefore was not rejected by this test.

The fitted quadratic curvature was

$$\beta_2 = -0.2294 \pm 0.1293,$$

with bootstrap 95% interval

$$[-0.4849,\ 0.0254]$$

and two-sided $p = 0.076$. The evidence for curvature is consequently strong and reproducible but still falls short of the conventional $p < 0.05$ criterion.

5 What circuit depth implies for model choice

There is no universal gate-depth boundary at which one should switch from a linear to a quadratic or exponential extrapolator. The relevant control variable is closer to an accumulated circuit error or "circuit area," which depends jointly on depth, two-qubit error rates, gate composition, observable support, coherence, and the fidelity with which folding scales those errors.

Majumdar *et al.* varied two-qubit depth and depolarizing error and constructed calibration phase diagrams for linear, quadratic, exponential, and "no-fit" regions [2]. Their results support the following qualitative progression:

1. At low depth and low accumulated error, the response is locally linear; linear fitting is stable and has the smallest variance.

2. At greater circuit area, measurable curvature develops and quadratic or exponential fits can lower RMSE.

3. At still larger depth/error, amplified expectation values approach noise saturation. Linear extrapolation can be precisely wrong, while exponential fitting can become unstable; no fitted ZNE model is then reliable.

For factors $\{1, 3, 5\}$, the same work found curved extrapolators increasingly useful at larger depths, including hardware tests out to two-qubit depth 70. With much smaller factors such as $\{1, 1.1, 1.2\}$, linear fitting remained useful farther into the deeper/noisier regime and the no-fit region shrank [2]. This is consistent with using the present fractional grid rather than factors 3 and 5.

Exponential behavior has a physical motivation under Markovian or Pauli noise, for which Pauli expectations can decay as one or several exponentials [3, 4]. Experiments at large scale have also reported that exponential extrapolation can outperform linear fitting when changes among amplified-noise estimates are well resolved, while downgrading the model when the exponential

becomes unstable [5]. These results do not make exponential fitting universally correct: coherent, non-Markovian, gate-dependent, leakage, and nonuniformly amplified errors can violate the assumed response.

For the present entry, the observed flattening, improved curved-model bias, and held-out prediction are consistent with the "moderate accumulated error" regime in which literature finds curved models useful. The final selection is nevertheless not decisive: the exponential has the smallest observed simulator RMSE, the quadratic has almost the same fit quality with smaller variance, and the linear fit remains statistically acceptable.

6 Recommended use

1. Retain all factor estimates and their total variances, including folding-pattern variance.
2. Use predeclared pilot-derived weights; do not select weights or models merely because they land closest to a target that would be unknown on hardware.
3. Report the linear estimate as the stable low-variance reference. Report quadratic and offset-exponential estimates as curved-model alternatives for this deep entry.
4. Prefer predictive checks, bootstrap intervals, and residual diagnostics over choosing a model from circuit depth alone.
5. Reject ZNE when amplified points have saturated or when fitted intercepts are unstable under modest changes of factors/model.
6. Before hardware deployment, fold the complete physical circuit, including measurement-basis rotations, or explicitly quantify the unamplified error contribution.

7 Saved artifacts

The final independent-seed pooled comparison is stored in

```
saved/runtime_runs/
  20260724T024719Z_h2o_1.0_state0_orbital0_split_v1_qwc_fake_quebec/
  probabilistic_zne_entry_1_1_20260727T162129Z/
```

The directory includes the source run result, calibrated readout data, raw measurement histograms, folded circuits, numerical ZNE arrays, `zne_model_comparison.json.gz`, and `zne_model_bootstrap_samples.npz`. A companion independently seeded high-shot run is

```
probabilistic_zne_entry_1_1_20260727T161644Z/
```

References

[1] T. Giurgica-Tiron, Y. Hindy, R. LaRose, A. Mari, and W. J. Zeng, "Digital zero noise extrapolation for quantum error mitigation," *2020 IEEE International Conference on Quantum Computing and Engineering*, pp. 306–316 (2020). doi:10.1109/QCE49297.2020.00045.

[2] R. Majumdar, P. Rivero, F. Metz, A. Hasan, and D. S. Wang, "Best practices for quantum error mitigation with digital zero-noise extrapolation," *2023 IEEE International Conference on Quantum Computing and Engineering*, pp. 881–887 (2023). doi:10.1109/QCE57702.2023.00102.

[3] Z. Cai, "Multi-exponential error extrapolation and combining error mitigation techniques for NISQ applications," *npj Quantum Information* **7**, 80 (2021). doi:10.1038/s41534-021-00404-3.

[4] M. Krebsbach, B. Trauzettel, and A. Calzona, "Optimization of Richardson extrapolation for quantum error mitigation," arXiv:2201.08080 (2022). https://arxiv.org/abs/2201.08080.

[5] Y. Kim *et al.*, "Evidence for the utility of quantum computing before fault tolerance," *Nature* **618**, 500–505 (2023). doi:10.1038/s41586-023-06096-3.